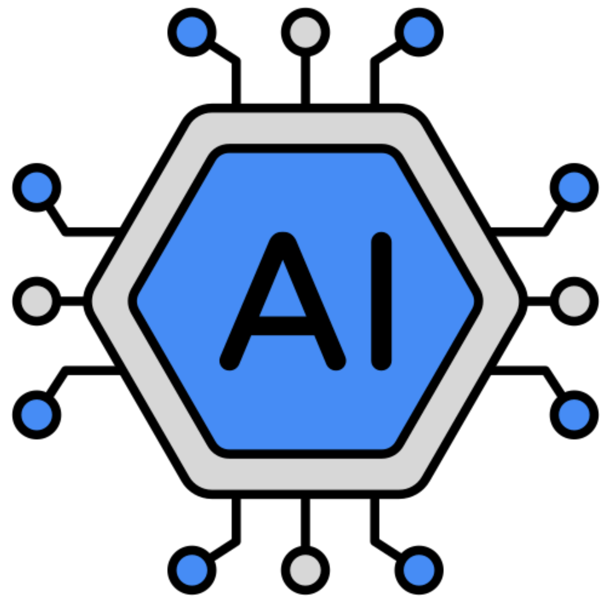
[Lab4]
General Prompt vs. Step-by-Step Instructions –A/B Comparison Lab

Learning Goal

Toexperiencetheimpactof logicalsequencingbycomparinga'GeneralPrompt' with 'Step-by-Step Instructions'

Task Overview

- Lab 4-1: Understanding the Task and Scenario (10 min)
- Lab 4-2: Designing and Executing Prompt A (10 min)
- Lab 4-3: Designing and Executing Prompt B (15 min)
- Lab 4-4: Comparing and Summarizing Results (10 min)
- Lab 4-5: Summarizing A/B Usage Criteria and Creating My Own Rule (10min)



Lab 4-1: Understanding the Task and Scenario (10 min)

Mission

Analyze a task scenario that requires multiple logical stages to identify why a single instruction might fail.

Discussion Point

Deconstruct the task into its "Hidden Steps" before you start prompting.

Points

1. What are the 3-4 distinct logical steps the AI must take to complete this task successfully?
2. If we give all instructions at once, at which specific step is the AI most likely to make a calculation or reasoning error?
3. Does Step 2 of this task depend on the result of Step 1? How does this influence your prompt design?

Activity

Identify the "Step-by-Step" sequence for this scenario:

"To solve this task perfectly, the AI must first get the available hours and days, then modify the plan according to breaks and unpredictable situation, and finally gives the right plan both A and B types for do not miss any high priority steps."

Lab 4-2: Designing and Executing Prompt A (10 min)

Mission

Design and execute a "General Technique" prompt by providing all instructions in a single, unified block to test the AI's baseline performance.

Discussion Point

Observe the risks of "Information Overload" in a single instruction.

Points

1. When you pack multiple requirements into one paragraph, how do you expect the AI to prioritize them?
2. In a general prompt, the AI gives you the final result immediately. Do you know how it calculated the study hours or breaks?
3. Which specific detail from the scenario is most likely to be "forgotten" by the AI in this single-step approach?

Activity

Execute Prompt A and evaluate the immediate output:

"Prompt A produced a result that felt unrealistic and demotivating because the AI missed the specific constraint of clearly mapping out the break times and assumed I wanted all hard tasks first.

Lab 4-3: Designing and Executing Prompt B (15 min)

Mission

Design and execute an "Advanced Technique" prompt using Step-by-Step instructions to guide the AI's reasoning process.

Discussion Point

Guide the AI's "Chain of Thought" by breaking the task into logical milestones.

Points

1. Instead of asking for a "Plan," try asking the AI to step-by-step.
2. How does forcing the AI to finish "Step 1" before moving to "Step 2" reduce the chance of mathematical errors?
3. How does seeing the intermediate steps help you verify if the AI's "Reasoning" is actually correct?

Activity

Execute Prompt B and compare the "Depth of Logic" with Prompt A:
"By using Step-by-Step instructions, the AI was able to show exact break times and put easy tasks first for motivation, which was missing in the general approach of Prompt A."

Lab 4-4: Comparing andSummarizingResults (10 min)

Mission

Perform a side-by-side analysis of the outputs from promptsto evaluate which technique delivers higher professional accuracy.

Discussion Point

Critically analyze the "Control Gap" between single and sequential instructions.

Points

1. Which prompt followed the complex constraints without any errors?
2. In which version was it easier to verify if the AI's internal logic was actually correct?
3. If you run both prompts again, which one do you expect will produce a more consistent and reliable result?

Activity

Share your "Performance Comparison" with the class:
"The biggest difference was the details of the schedule. While Prompt A felt like a lazy guess,
Prompt B provided a useful and realistic solution because it forced to the AI to calculate times step by step."

Lab 4-5: Summarizing A/B Usage Criteria & My Own Rule (10 min)

Mission

Synthesize your findings to define exactly when to use "General" vs. "Step-by-Step" prompts in your future workflows.

Discussion Point

Reflect on the following criteria to finalize your "Golden Rule."

Points

1. **Prompt A (General):** Suitable for low-stakes, simple requests, or quick brainstorming where format doesn't matter.
2. **Prompt B (Step-by-Step):** Mandatory for multi-stage logic, math/data accuracy, or professional-grade deliverables.

Activity

Share your "ExecutionRule" for choosing a prompt style:

"I will use Prompt A when I just need quick and simple ideas, and Prompt B when I need personal and complicated plan to ensure my AI outcomes are always OK for real part-time work life."